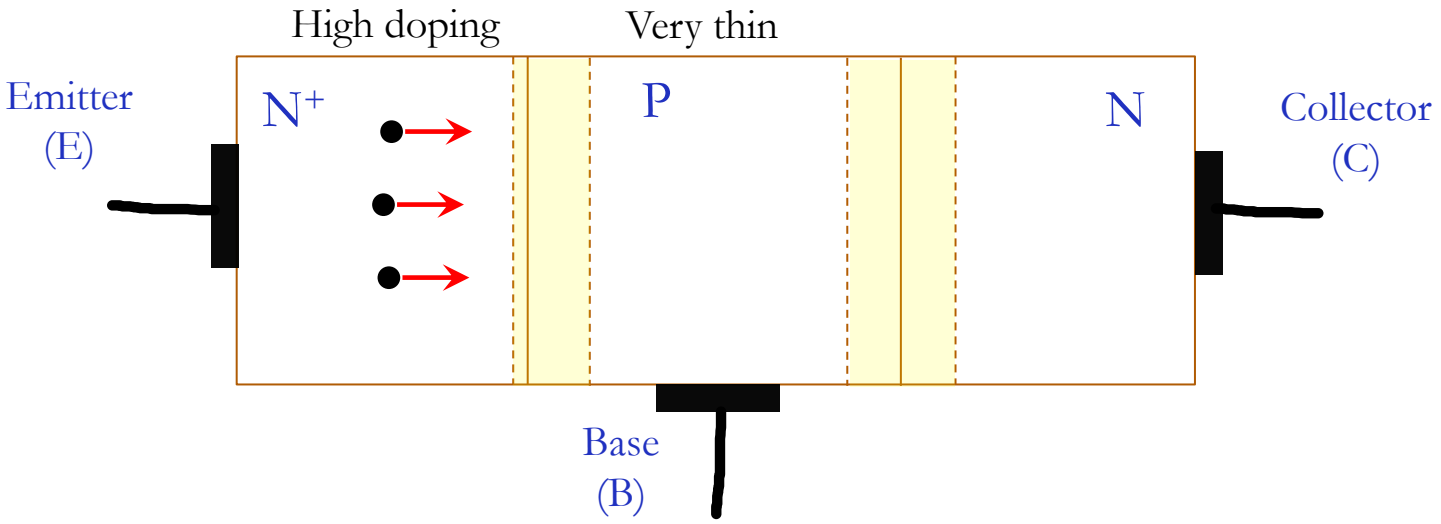
Bipolar Junction Transistors

- BJT Basic Structure.
- Current in BJTs and Operation Modes.
- DC Analysis (Biasing).

Device Structure

The bipolar junction transistor is basically two diodes (pn junctions) connected back-to-back.

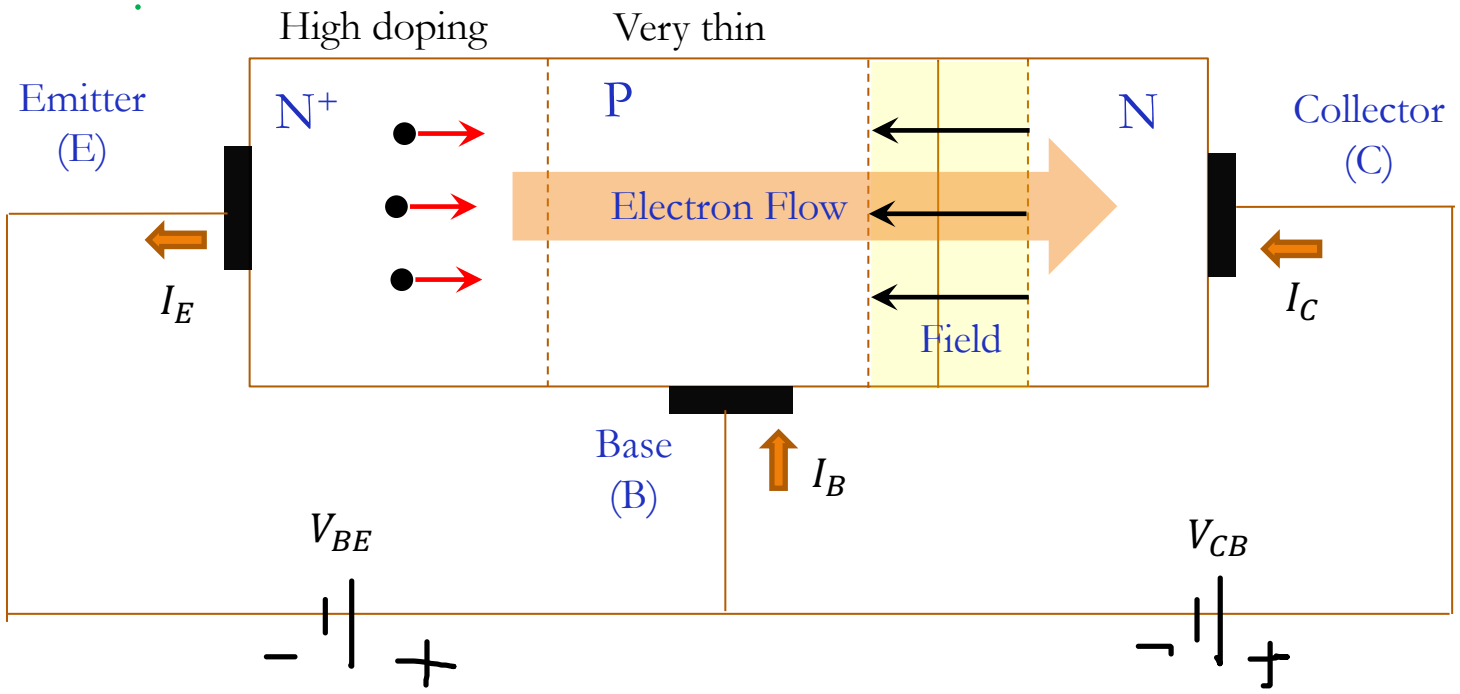


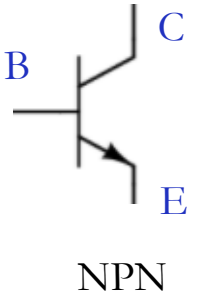
Four possible scenarios for current.

1) The two junction are reversed biased (cutoff mode).
2) BE junction forward, BC junction reversed (active mode).
3) BE junction reversed, BC junction forward (reverse active mode).
4) The two junction are forward biased (saturation mode).

The Active Mode

The active mode is the mode of operation we use in amplifier circuit.





Current relations

$V_{BE} = 0.7\ V$

$I_E = I_B + I_C$

$I_E = (\beta + 1) I_B$

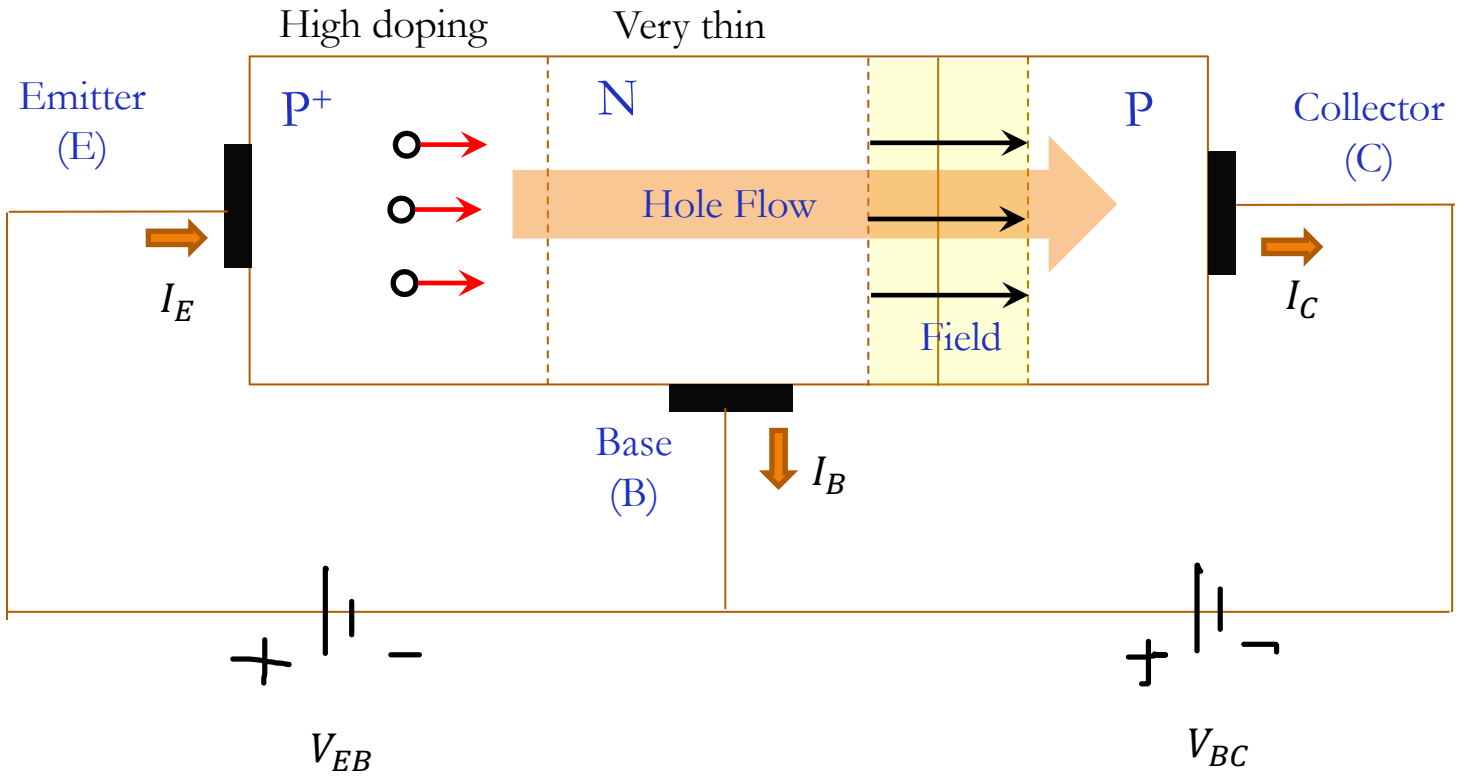
$\beta = \frac{I_C}{I_B}$

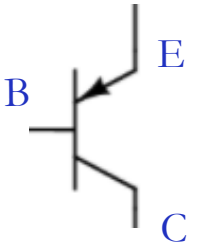
$\alpha = \frac{I_C}{I_E}$

$\alpha = \frac{\beta}{\beta + 1}$

The Active Mode

The bipolar junction transistor is basically two diodes (pn junctions) connected back-to-back.





PNP

Current relations

$$V_{EB} = 0.7\ V$$

$$I_E = I_B + I_C$$

$$I_E = (\beta + 1) I_B$$

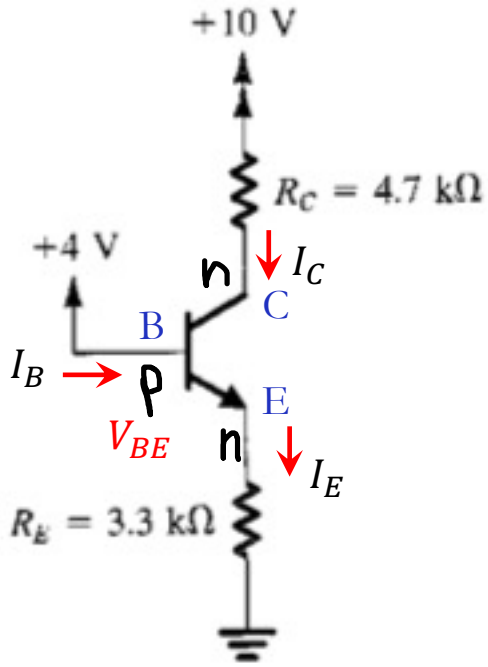
$$\beta = \frac{I_C}{I_B}$$

$$\alpha = \frac{I_C}{I_E}$$

$$\alpha = \frac{\beta}{\beta + 1}$$

Examples

1) Analyze the DC BJT circuit below. (β=100).



Assume active mode

$$V_{BE} = 0.7\,V = V_B - V_E$$

$$V_B = 4\,V$$

$$V_E = 4 - 0.7 = 3.3\,V$$

$$I_E = 1\,mA$$

$$I_C = \frac{\beta}{\beta+1} I_E = \frac{100}{101} 1\,mA = 0.99\,mA$$

$$I_B = I_E - I_C = 10\,\mu A$$

$$V_C = 10 - I_C R_C = 5.35\,V$$

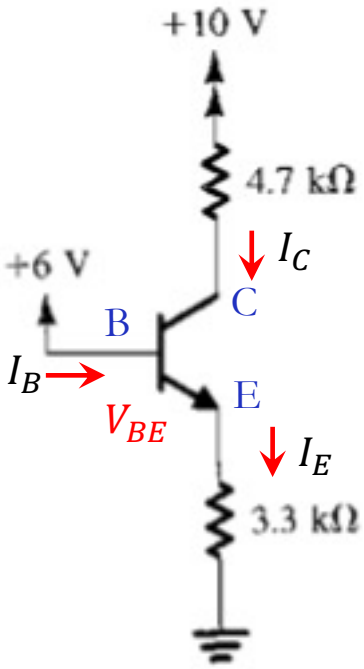
$$V_C > V_B$$

CB junction is reversed biased

Assumption is correct

Examples

2) Analyze the DC BJT circuit below. (β=100).



Assume active mode

$$V_{BE} = 0.7\ V = V_B - V_E$$

$$V_B = 6\ V$$

$$V_E = 6 - 0.7 = 5.3\ V$$

$$I_E = 1.61\ mA$$

$$I_C = \frac{\beta}{\beta+1} I_E = \frac{100}{101} 1.61\ mA = 1.59\ mA$$

$$I_B = I_E - I_C = 20\ \mu A$$

$$V_C = 10 - I_C R_C = 2.53\ V$$

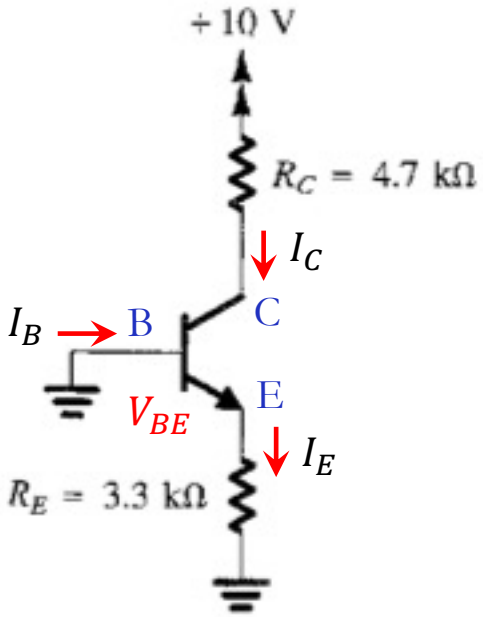
$$V_C < V_B$$

CB junction is forward biased

Not active mode!

Examples

3) Analyze the DC BJT circuit below. (β=100).



Assume active mode

$$V_{BE} = 0.7\,V = V_B - V_E$$

$$V_B = 0\,V$$

$$V_E = 0 - 0.7 = -0.7\,V$$

$$I_E = -0.21\,mA$$

Not possible. BE junction is cutoff

$$I_E = 0$$

$$I_C = 0$$

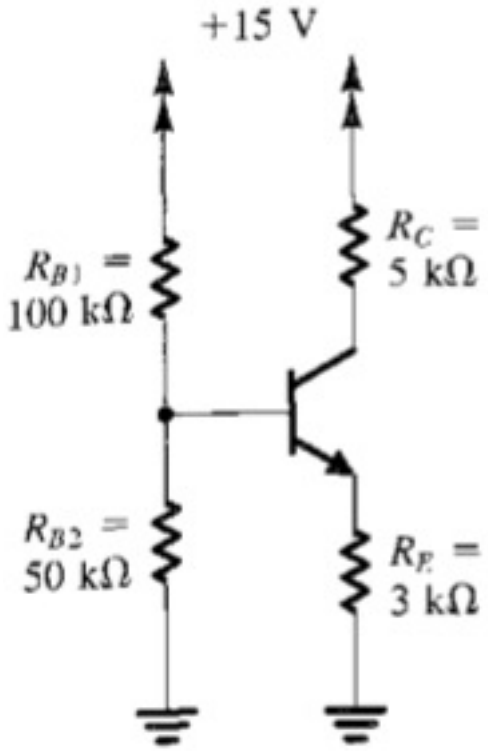
$$V_C = 10\,V$$

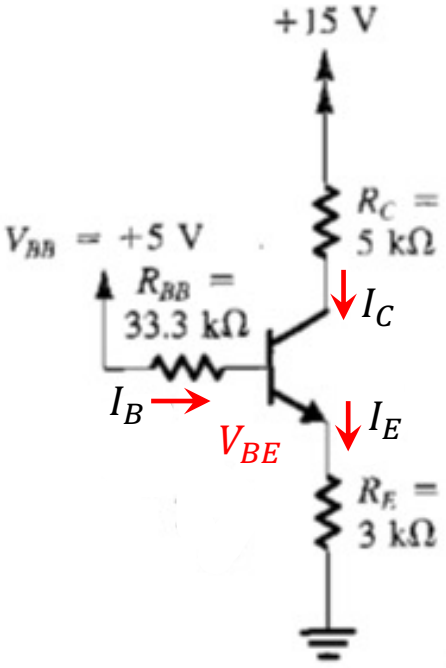
$$V_C > V_B$$

CB junction is reversed biased

Examples

4) Analyze the DC BJT circuit below. (β=100).





Assume active mode

$$V_{BE} = 0.7\ V$$

KVL in the B-E circuit

$$5 - I_B R_{BB} - V_{BE} - I_E R_E = 0$$

$$5 - I_B(33.3) - 0.7 - (\beta + 1) I_B(3) = 0$$

$$I_B = 12.8\ \mu A$$

$$I_C = \beta I_B = 100(12.8\mu) = 1.28\ mA$$

$$I_E = I_B + I_C = 1.29\ mA$$

$$V_C = 15 - I_C R_C = 8.6\ V$$

$$V_B = 5 - I_B R_{BB} = 4.57\ V$$

$$V_C > V_B$$

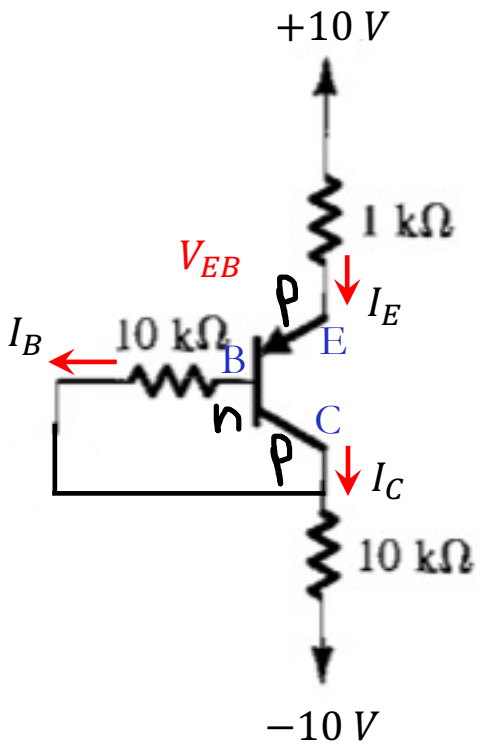
CB junction is reversed biased

Assumption is correct

$$V_E = I_E R_E = 3.87\ V$$

Examples

5) Analyze the DC BJT circuit below. (β=50).



Assume active mode

$$V_{EB} = 0.7\ V$$

KVL in the (+10) to (-10) branch

$$10 - I_E(1\ k) - V_{EB} - \frac{I_E}{\beta + 1}(10\ k) - I_E(10\ k) + 10 = 0$$

$$I_E = 1.72\ mA$$

$$V_E = 10 - (1.72\ m)(1\ k) = 8.28\ V$$

$$V_B = V_E - V_{EB} = 7.58\ V$$

$$V_C = (1.72\ m)(10\ k) - 10 = 7.2\ V$$

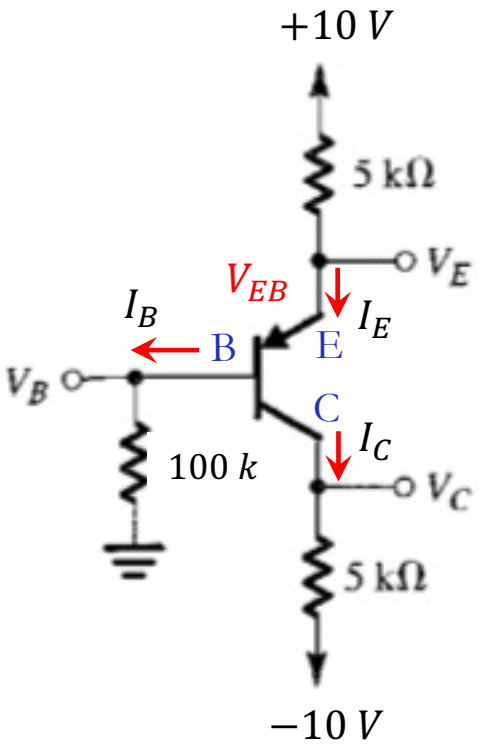
$$V_C < V_B$$

CB junction is reversed biased

Assumption is correct

Examples

6) Find the parameters α and β of the transistor below given one measurement.



$$V_B = +1\ V$$

$$V_E = +1.7\ V$$

$$I_E = \frac{10 - 1.7}{5\ k} = 1.66\ mA$$

$$I_B = \frac{1}{100\ k} = 10\ \mu A$$

$$\beta = \frac{1.66\ mA}{10\ \mu} - 1 = 165$$

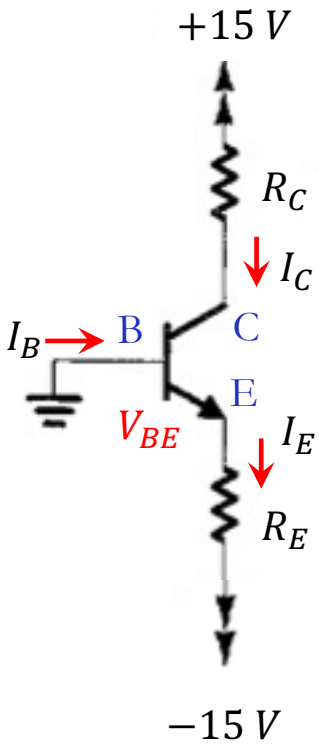
$$\alpha = \frac{165}{165 + 1} = 0.994$$

$$V_C = 0.994(1.66\ m)(5\ k) - 10 = -1.75\ V < V_B$$

Active mode

Examples

7) Design the circuit for I_C=2 mA and V_C=5 V. (β=100).



As given, the circuit is in active mode.

$$V_C = 5\ V$$

$$V_B = 0\ V$$

$$V_E = 0 - 0.7 = -0.7\ V$$

$$I_C = 2\ mA \rightarrow R_C = \frac{15-5}{2\ m} = 5\ k\Omega$$

$$I_E = \alpha\ I_C = \frac{100}{101}\ 2\ m = 2.02\ mA$$

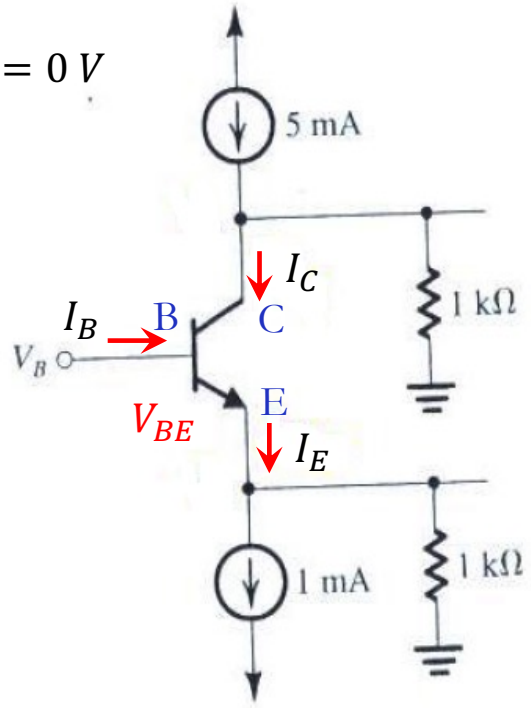
$$R_E = \frac{-0.7-(-15)}{2.02\ m} = 7.07\ k\Omega$$

Examples

8) Analyze the DC BJT circuit below. (β=100).

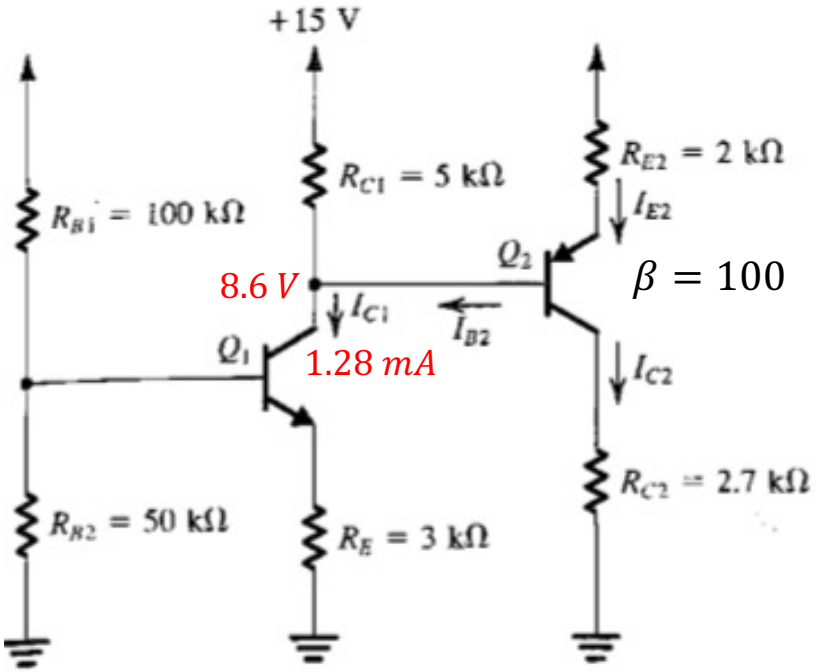
$V_B = 1\ V$

$V_B = 0\ V$



Examples

9) Another transistor stage is added to example (4).



$\beta = 100$

$$V_{B2} = V_{C1} = 8.6\ V$$

$$V_{E2} = 8.6 + 0.7 = 9.3\ V$$

$$I_{E2} = \frac{15 - 9.3}{2\ k} = 2.85\ mA$$

$$I_{C2} = 2.82\ mA$$

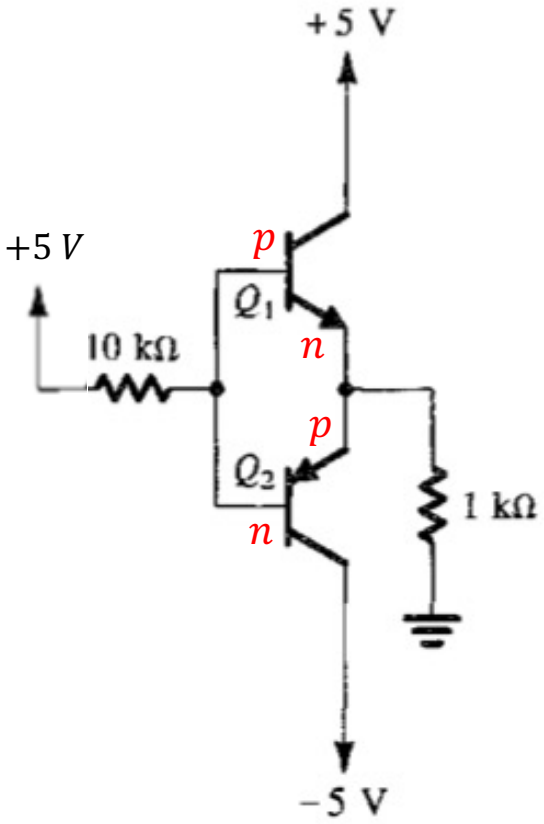
$$V_{C2} = (2.82\ m)(2.7\ k) = 7.62\ V$$

Active mode

$$I_{B2} = 28.2\ \mu A$$

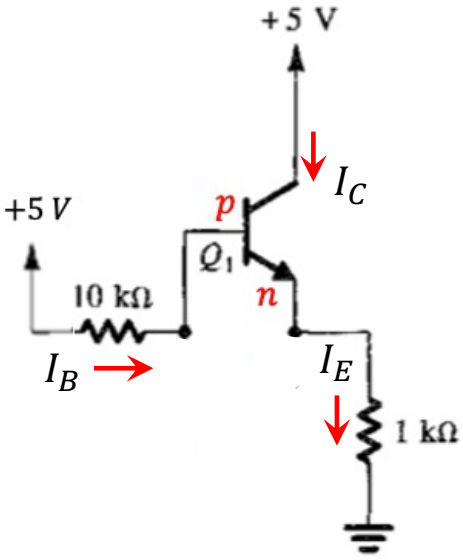
Examples

10) Analyze the DC BJT circuit below. (β=100).



The two transistors cannot be both ON, because the EB junctions are connected across each other.

With the given biasing, Q_2 is OFF.



Assume active mode

$$V_{BE} = 0.7\ V$$

KVL in the B-E circuit

$$5 - I_B(10\ k) - 0.7 - (\beta + 1)I_B(1\ k) = 0$$

$$I_B = 38.7\ \mu A$$

$$I_C = 3.87\ mA$$

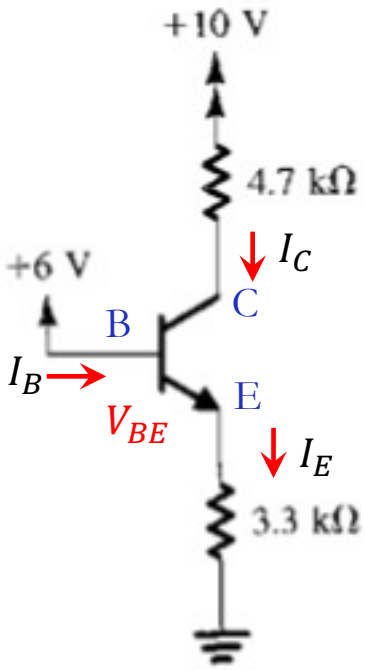
$$I_E = 3.91\ mA$$

$$V_B = 5 - (38.7\ \mu)(10\ k) = 4.6\ V < V_C$$

Active mode

Examples: Saturation Mode

11) Analyze the DC BJT circuit below. (β=100).



Circuit was checked in a previous example. It was not in active mode.

$$V_{BE} = 0.7\ V = V_B - V_E$$

$$V_B = 6\ V$$

$$V_E = 6 - 0.7 = 5.3\ V$$

$$I_E = 1.61\ mA$$

The BC junction is also forward. Then,

$$V_{BC} = 0.5\ V = V_B - V_C$$

$$V_C = 5.5\ V$$

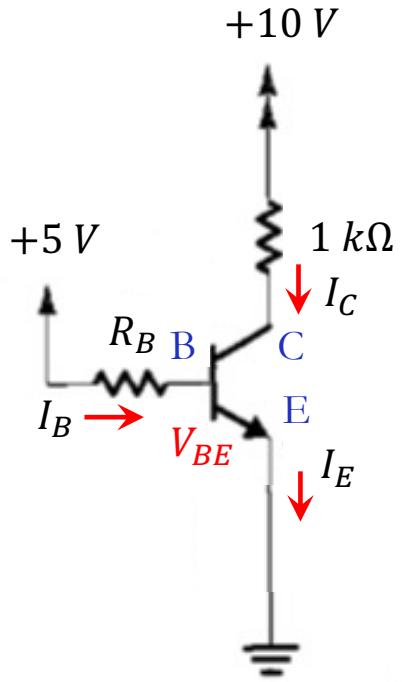
$$I_C = \frac{10 - 5.5}{4.7\ k} = 0.96\ mA$$

$$I_B = I_E - I_C = (1.61 - 0.96) = 0.65\ mA$$

$$\beta_{forced} = I_C/I_B = 1.48$$

Examples: Saturation Mode

12) Design the circuit for deep saturation with overdrive factor of 10. (β=50-100).



Saturation means both junctions are forward

$$V_E = 0\ V$$

$$V_B = 0.7\ V$$

$$V_C = 0.2\ V$$

Use the minimum β for the design.

$$I_C = \frac{10 - 0.2}{1\ k} = 9.8\ mA$$

$$I_B = \frac{I_C}{\beta_{min}} = \frac{9.8\ m}{50} = 0.196\ mA$$

$$I_{B_sat} = 10 \times 0.196\ m = 1.96\ mA$$

$$R_B = \frac{5 - 0.7}{1.96\ m} = 2.2\ k\Omega$$

Q & A

Send me an email if
you have any
questions.